Global kinetic design for rare completion

A direct scalar experimental oracle, a continuum optimizer, and its physical boundary

Independent Astra construction

5 November 2026

Abstract

We construct a global optimization operation for a rare desired first event. The design variables are certified reciprocal kinetic multipliers which preserve one unknown invariant distribution. The state space may be exponential in the number of local variables. The construction chooses a continuous allocation of the multipliers whose mean completion time from equilibrium conditioned on not yet completing is within a prescribed multiplicative factor of the global optimum. It uses local operating experiments, a priced conditional-target sampler, gap certificates, and ordinary linear optimization over the control constraints. It does not acquire a full trajectory law or simulate an entire rare waiting time. A direct short reverse no-return statistic estimates the normalized mean; the common stationary target probability cancels from the optimization. The normalized objective has a rarity-independent curvature bound, so finite inexact Frank–Wolfe optimization needs polynomially many scalar queries, rather than a grid exponential in the number of controls. The kinetic multiplier interface is an explicit extra intervention premise; it is not inferred from arbitrary field-dependent clocks. This is a theoretical design capability, not a completed economical physical catalyst. Bounded multipliers can improve mean completion time by at most their largest allowed gain, so the actual rare completion exposure remains.

1 The selected independent question

Can local experiments select the globally best continuous allocation of reversible kinetic acceleration, for a specified rare completion event, without learning every operating rate, reconstructing the entire global Markov chain, or generating complete first-event records?

The direction of the objective is essential. This document minimizes mean time to a *desired* event. It does not turn the maximization problem of preventing a catastrophe into a convex problem. A finite menu of preventive fields is a different previously earned operation. Nor is minimum mean completion the same objective as minimum Kemeny constant, fastest mixing, or minimum stationary target probability.

The result combines a new explicit interface with established machinery: matrix-fractional convexity and reversible potential theory; the preserved short no-return spectral statistic; finite experimental static and operating updates; and inexact conditional-gradient optimization. Historical priority for that combination is not established.

2 Mathematical and experimental access

Let Ω be a finite, succinctly represented state space and let $B \neq \varnothing$ be the known target. Write $A = \Omega \setminus B$, $p = \pi(B)$ and $a = 1 - p$. Assume $0 < p \leq 1/2$. There are d local generators Q_j. Each is reversible with respect to the same full-support invariant law π. The admissible design family is

$$Q(c) = \sum_{j=1}^{d} c_j Q_j, \qquad c \in \mathcal{C} \subseteq [1, C]^d, \quad C \geq 1. \tag{1}$$

$\mathcal{C}$ is a nonempty compact rational polytope with an explicitly priced linear minimization operation. A budget polytope such as $\sum_j w_j(c_j - 1) \le b$ with box constraints is an example. The baseline $\sum_j Q_j$ is irreducible. The number of channels, their local supports, and the control constraints are input size; the number of global states need not be polynomial.

Choose one rational

$$R \ge 2(C+1)\sum_j r_j, \qquad r_j \ge \max_x[-Q_j(x,x)], \qquad P(c) = I + Q(c)/R.$$

The slightly enlarged numerical domain $[1, C+1]^d$ is only used for one-sided finite differences. It requires no extra physical deployment amplitude: its hypothetical updates can be composed from baseline experimental channels as described below. Every $P(c)$ on this enlarged domain is lazy and positive semidefinite. A certified normalized full-chain gap $g \in (0, 1]$ holds uniformly there. The baseline gap suffices since each added Dirichlet form is positive semidefinite.

The operation is conditional on the following earned or supplied services.

1. A finite sampler for $\pi(\cdot \mid B)$ and, for the nonrare branch, for $\pi(\cdot \mid A)$, with specified conditional total-variation error and work. A relative count of p is needed only when a supplied upper bound does not establish the rare branch. The accepted correlated Gibbs static construction supplies these at its original contrast, pinned mixing, and local-degree scope. A singleton B needs no conditional target sampling; a certified simple upper bound on p can avoid counting.

2. A requested-context baseline operating update of each Q_j, obtained from faithful local copies with priced state pins, context clamps, actual operating fields, rate bounds, detector and timer guarantees. The preserved experiment-fed operating-coin construction supplies such an update with a uniform conditional error. It does not transport a clock from a balancing field to the operating field.

3. Structural closure, local Markov behavior, the common reversible invariant, known supports, and numerical mixing/static margins. They are premises. Local observations do not prove absence of hidden channels.

4. For *deployment*, an actual reciprocal multiplier service implementing (1), or a declared uniform multiplicative conductance envelope around it. Arbitrary field changes, whose unknown attempt clocks can depend on the field, do not supply this service. Its response law, calibration, precision, slew, work and setup costs must be supplied. Without it the conclusion is a design theorem for an accessible generator family, not an intervention on the original object.

There is an explicit way to generate hypothetical updates for any requested $c \in [1, C+1]^d$ without relearning kinetics. Choose j with weight $r_j / \sum_i r_i$, retain the choice with probability $c_j/(C+1)$, and, if retained, apply $I + Q_j/(2r_j)$. The resulting kernel is $P(c)$ when $R = 2(C+1)\sum_j r_j$; larger R adds a known null probability. Finite rational selectors and multiplier coins use an assigned error budget. The local operating factory implements the requested baseline channel. This is algebra on the observed baseline generator, not an assumption about an unmeasured bias clock.

If the static generator is an unknown pairwise Gibbs model, balanced local experiments supply the accepted static-weight and pinned-sampling services. Their field metrology and repeated-cell conditional-error premises remain unchanged. In particular, a sparse energy graph does not prove sparse arbitrary kinetic context tables; this route queries contexts as needed instead of enumerating those tables.

3 The normalized objective and its geometry

Let τ_B denote the physical first hitting time and define

$$M(c) = \mathbb{E}_{\pi(\cdot|A),Q(c)}\tau_B, \qquad f(c) = pRM(c).$$

The factor pR is common to every design. Minimizing f and M is therefore exactly the same decision; no estimate of their astronomical absolute scale is needed to interpret an optimizer.

Let $K(c) = -Q_A(c)/R$, where the subscript denotes killing on B, and $v = \mathbf{1}_A$, in $\ell^2(A,\pi)$. Put $K_j = -Q_{j,A}/R$. Then

$$f(c) = \frac{p}{a}\langle v, K(c)^{-1}v\rangle_\pi, \qquad K(c) = \sum_j c_j K_j, \qquad K_j \succeq 0. \tag{2}$$

All derivatives below are mathematical identities; the algorithm does not form these exponentially large matrices.

Theorem 1 (Rarity-free objective geometry)**.** *For every $c \in [1, C+1]^d$,*

$$\frac{1}{2} \le f(c) \le \frac{1}{g}, \tag{3}$$

$$\partial_j f(c) \le 0, \qquad \sum_j c_j[-\partial_j f(c)] = f(c), \tag{4}$$

$$0 \le D^2 f(c)[h,h] \le \frac{2}{g}\|h\|_\infty^2. \tag{5}$$

In particular f is convex, $\|\nabla f(c)\|_1 \le 1/g$, and its Frank–Wolfe curvature on $\mathcal{C}$ is at most $2(C-1)^2/g$.

Proof. Zero extension of safe functions and Poincare give $K(c) \succeq gpI$. Hence $\langle v, K^{-1}v\rangle \le a/(gp)$, proving the upper bound. Let $j_c = \langle v, K(c)v\rangle$ be stationary boundary flux in normalized units. Cauchy–Schwarz gives $\langle v, K^{-1}v\rangle \ge a^2/j_c$. Reversibility gives $j_c \le p$ (in fact laziness gives $j_c \le p/2$), so $f \ge a \ge 1/2$.

For $u = K^{-1}v$, differentiation gives

$$\partial_j f = -\frac{p}{a}\langle u, K_j u\rangle, \qquad D^2 f[h,h] = \frac{2p}{a}\langle K(h)u, K^{-1}K(h)u\rangle, \quad K(h) = \sum_j h_j K_j.$$

These prove convexity, negativity of coordinate derivatives, and (4). Since $c_j \ge 1$,

$$-\|h\|_\infty K(c) \preceq K(h) \preceq \|h\|_\infty K(c).$$

Thus the self-adjoint operator $K^{-1/2}K(h)K^{-1/2}$ has norm at most $\|h\|_\infty$. Substitution into the second derivative gives $D^2 f[h,h] \le 2\|h\|_\infty^2 f(c) \le 2\|h\|_\infty^2/g$. For a chord $h = s - c$ in the control box, $\|h\|_\infty \le C - 1$. Taylor's integral remainder gives the stated curvature. □

The ℓ^∞ geometry in (5) matters: the curvature does not contain a sum over the number of global states, or an inverse p. Control dimension still enters the cost of estimating a gradient.

4 A direct scalar oracle with fixed update counts

Here the oracle returns $\widetilde{f}(c)$ satisfying $|\widetilde{f}(c) - f(c)| \le \nu$ with conditional confidence $1 - \delta_v$, where $0 < \nu \le 1$. It returns no event time, terminal state or precursor record. It does not construct the entire CDF. No global continuous event clock is required: the mean of the continuous uniformized hitting time equals the mean number of potential updates divided by R.

4.1 One common branch for every control

Set

$$\xi = \frac{\nu g}{64}, \quad L = \left\lceil \log_2 \frac{4}{g\xi} \right\rceil, k = \lceil L/g \rceil, \quad p_{\text{gate}} = \frac{\xi}{2k}. \tag{6}$$

All ceilings are implementable by rational comparisons with powers of two. Use a certified upper bound $p_U \le p_{\text{gate}}$ if available. Otherwise obtain a relative static interval $[p_L, p_U]$ with ratio at most two. If $p_U \le p_{\text{gate}}$ use the rare branch. Otherwise $p_L > p_{\text{gate}}/2$ and use the direct branch. The branch is shared by all optimization queries because the invariant law is common.

4.2 Rare branch: escape briefly from a deliberately supplied target

Draw $Y \sim \pi(\cdot \mid B)$. Apply one $P(c)$ update. Return $Z = 1$ exactly when this update leaves B and the next k updates stay in A. Let $z(c) = \mathbb{E}Z$. This costs $k+1$ potential updates, even when the true completion mean is exponentially large.

Lemma 2 (Short scalar identity and approximation)**.** *Under $p \le p_{\text{gate}}$,*

$$z(c) \ge g/4, \qquad \left| f(c) - \frac{1}{z(c)} \right| \le \nu/4.$$

Proof. The preserved killed spectral facts give eigenvalues $0 < \lambda = \lambda_1 \le \lambda_2 \le \cdots \le 1$, $\lambda \in [gp, p/a]$, $\lambda_2 \ge g$, weights $\alpha_i \ge 0$ summing to one, and $r = 1 - \alpha_1 \le p/(ag)$. Reversibility gives the exact identity

$$pz = a \sum_i \alpha_i \lambda_i (1-\lambda_i)^k, \qquad f = p \sum_i \frac{\alpha_i}{\lambda_i}.$$

Our gate implies $k\lambda \le \xi$ and $r \le \xi$. The slow term divided by a is at least $(1-\xi)^2\lambda$. The sum of fast terms, divided by $a\lambda$, is at most $e^{-gk}/(ag) \le \xi/2$. Consequently

$$(1-2\xi)\lambda \le pz/a \le (1+\xi)\lambda, \qquad 1-\xi \le (f/p)\lambda \le 1.$$

Thus $z \ge a(1-2\xi)g \ge g/4$ and $(1-3\xi)a/z \le f \le (1+\xi)a/z$. Since $a/z \le 4/g$ and $p \le \xi$, $|f - 1/z| \le 12\xi/g + 4p/g \le 16\xi/g = \nu/4$. □

Take $t = \nu g^2/128$. Make each implemented trial's conditional mean differ from the ideal $z(c)$ by at most $t/2$, for example using a conditional-target draw at TV error $t/4$ and each of $k+1$ updates at error $t/[4(k+1)]$. Use

$$n_z = \left\lceil \frac{2}{t^2} \log \frac{2}{\delta_v} \right\rceil$$

trials. Martingale Hoeffding, applied after conditioning on the previous history, gives $|\bar{Z} - z| \le t$ with probability $1 - \delta_v$. No independence of repeated physical cells is asserted beyond their declared conditional kernel guarantees. On this event $\bar{Z} \ge g/8$ and

$$\left| \frac{1}{\bar{Z}} - \frac{1}{z} \right| \le 32t/g^2 = \nu/4.$$

Return an outward-rounded reciprocal, with rounding error at most $\nu/4$. A zero or invalid empirical denominator returns a declared failure value; it is outside the simultaneous good event. The result is within ν. The conservative statistical count is $O(\nu^{-2} g^{-4} \log(1/\delta_v))$; it is not claimed optimal.

4.3 Nonrare branch: a bounded direct mean

Refine the single shared static count so $|\widehat{p}/p - 1| \le \rho = \nu g/64$, preserving a ratio-two interval. Set

$$L_d = \left\lceil \log_2 \frac{16}{\nu g} \right\rceil, J = \left\lceil \frac{L_d}{g p_L} \right\rceil, W = p_U J \le \frac{2L_d}{g} + 1.$$

From π_A, let H_B be the number of $P(c)$ updates to first hit B. Estimate $\widehat{p}\mathbb{E}(H_B \wedge J)$ by bounded samples; replace W by $2W$ if the chosen $\widehat{p}$ lies just outside the count interval. The killed spectral bound gives

$$p\mathbb{E}(H_B - J)_+ \le g^{-1} e^{-gpJ} \le \nu/16.$$

The relative p error contributes at most $\rho/g \le \nu/64$. Choose each implemented trial's mean bias at most $\nu/16$, by giving its entire finite trajectory TV error at most $\nu/(16 \cdot 2W)$; allocate this to its initial draw and at most J updates. Conditional Hoeffding with $O(W^2\nu^{-2}\log(1/\delta_v))$ trials gives empirical error at most $\nu/4$. Rational output rounding has its own $\nu/4$ allowance. This also supplies the desired oracle.

There is no hidden numerical $1/p$ cost in this branch: $p_L > \xi/(4k)$ implies $J \le 4kL_d/(g\xi)+1$, a polynomial in g^{-1}, ν^{-1} and logarithms. Safe mass $a \ge 1/2$ makes the finite safe-start rejection cost bounded. All common static failures are charged once, separately from per-query conditional confidence.

5 Finite continuous design

Assume $0 < \varepsilon \le 1$ and put

$$\Delta = \varepsilon/2, \qquad W_c = \max(1, C-1), \qquad h = \frac{\Delta g}{32 d W_c}, \qquad \nu = \frac{\Delta h}{64 d W_c}.$$

Initialize at any rational $c_0 \in \mathcal{C}$. At iteration t obtain $d+1$ scalar values at c_t and $c_t + he_j$, and set

$$G_{t,j} = \frac{\widetilde{f}(c_t + he_j) - \widetilde{f}(c_t)}{h}.$$

Choose a feasible rational s_t satisfying

$$G_t \cdot s_t \le \min_{s \in \mathcal{C}} G_t \cdot s + \Delta/16,$$

and use $c_{t+1} = (1-\gamma_t)c_t + \gamma_t s_t$ with $\gamma_t = 2/(t+2)$. An explicit rational linear program supplies this operation in polynomial bit work; no exponentially sized list of designs is generated.

Theorem 3 (Global rare-completion allocation)**.** *Take*

$$T = \left\lceil \frac{16 W_c^2}{g\Delta} \right\rceil.$$

Give each of the $V = (d+1)T$ adaptive scalar calls conditional failure probability at most $\delta/(2V)$ and give common acquisition, static branching and margin certification total failure probability at most $\delta/2$. On the simultaneous good event, exact rational control arithmetic returns $c_T \in \mathcal{C}$ satisfying

$$M(c_T) \le (1+\varepsilon) \min_{c \in \mathcal{C}} M(c).$$

All numerical control/output rounding can instead be assigned additive objective allowance $\Delta/4$, with slack in the same conclusion. The number of scalar calls is $O(dW_c^2/(g\varepsilon))$; their required accuracy obeys $\nu^{-1} = O(d^2 W_c^2/(g\varepsilon^2))$.

Proof. Coordinate curvature gives

$$|G_{t,j} - \partial_j f(c_t)| \le h/g + 2\nu/h.$$

For any feasible competitor s, the inexact linear direction therefore satisfies

$$\nabla f(c_t)\cdot(s_t - c_t) \le \nabla f(c_t)\cdot(s - c_t) + D, \quad D \le dW_c(h/g + 2\nu/h) + \Delta/16 \le \Delta/8.$$

Convexity and curvature $C_f \le 2W_c^2/g$ imply, for $e_t = f(c_t) - \min f$,

$$e_{t+1} \le (1-\gamma_t)e_t + \gamma_t D + \tfrac{1}{2}\gamma_t^2 C_f.$$

The usual elementary induction, starting with $\gamma_0 = 1$, gives $e_T \le D + 2C_f/(T+2) \le 3\Delta/8$. This is less than Δ even after the declared rounding allowance. Since $\min f \ge 1/2$ and $\Delta = \varepsilon/2$, the additive bound implies $f(c_T) \le (1+\varepsilon)\min f$. The positive constant pR cancels. Conditional union bounds apply to the adaptive calls; their input points may depend on every earlier result. The required uniform conditional oracle guarantees were constructed above. □

A stopping certificate. At any iterate, let s_t be the approximate linear minimizer. If $\eta_G \ge \|G_t - \nabla f(c_t)\|_1$ is certified, then

$$0 \le f(c_t) - \min f \le G_t\cdot(c_t - s_t) + (C-1)\eta_G + \Delta/16.$$

The right side is an observable inexact Frank–Wolfe gap. It can stop an instance earlier than the sufficient worst-case iteration schedule, but its value is meaningful only with the same scalar-oracle confidence and finite arithmetic allowances.

Bit and memory costs. The controls, linear programs and value outputs are rational. Exact convex combinations have bit length polynomial in T and the input bit length. For finite arithmetic one can round to a feasible rational point with ℓ^∞ displacement η_c; the objective changes by at most η_c/g. A feasible rounding operation must be provided by the actual control polytope, not assumed for an arbitrary nonlinear constraint set. Working memory comprises a control vector, local state, counters, and the inherited static-sampling data structure. No vector of length $|\Omega|$ or long event record is needed.

6 Execution costs and a real physical deployment statement

Let $C_B(\theta)$, $C_A(\theta)$ and $C_P(\theta)$ denote the complete costs of a target draw, safe draw and operating update at the requested conditional error, respectively. These include all relevant physical experiments; they are not unit-cost oracles. In the rare branch one scalar call costs at most

$$n_z\left[C_B(t/4) + (k+1)C_P\left(\frac{t}{4(k+1)}\right) + C_{\text{predicate}} + C_{\text{bookkeeping}}\right].$$

In the nonrare branch it costs

$$O\big(W^2\nu^{-2}\log(1/\delta_v)\big)\left[C_A(\theta_A) + JC_P(\theta_P) + JC_{\text{predicate}} + C_{\text{bookkeeping}}\right],$$

where $\theta_A + J\theta_P \le \nu/(32W)$ is sufficient. Multiply by V. Add common static, acquisition and margin checks, linear optimization, output and deployment. This is an explicit polynomial reduction, not an economical numerical bill.

For the inherited one-coordinate operating factory, $C_P(\theta)$ contains $O(\log(1/\theta))$ priced races, each with a pin and release, $O(\rho^{-1}\log^2(1/\theta))$ capped dwell, and the declared timing resolution of order $\theta/[\rho\log(1/\theta)]$, with its conservative constants. Those costs may be large. This theorem changes what is estimated and the dimension of the design search; it does not claim that a physical race is cheap, or that its sufficient timer requirement is a universal lower bound. There is no cost of simulating an entire mean lifetime or constructing a full marked path. The model's actual completion runtime is still $M(c_T)$ and must be paid when the desired event is executed.

6.1 Finite preparation and finite slew

Suppose a full-system repairable implementation can run through target states during preparation, a state-mass floor $\zeta > 0$ and physical baseline gap $\gamma > 0$ are certified, and the state can be inspected at a declared completion time. Baseline evolution for

$$t_{\mathrm{mix}} \geq \gamma^{-1} \log(1/(\eta\zeta))$$

gives a pointwise density ratio in $[1-\eta, 1+\eta]$ relative to π from any start, using the standard finite-state L^2 to pointwise bound. Allow surplus for finite timing and gap calibration. Ramp from baseline to the chosen c_T with the actual finite slew. Every intermediate generator preserves π, so the same pointwise envelope is preserved by the time-inhomogeneous transition kernel during the ramp. This argument does not require a free instantaneous switch or a globally frozen network.

Inspect and retain a state in A. Since $a \geq 1/2$, one trial succeeds with probability at least $(1-\eta)/2$. On success the initial law ν_{c_T} obeys

$$e^{-\beta} \leq \frac{d\nu_{c_T}}{d\pi_A} \leq e^{\beta}, \qquad \beta = \log \frac{1+\eta}{1-\eta}.$$

Finite retries with this success floor give explicit logarithmic caps and preparation abort probability. The target must be nondestructive until the declared mission starts; irreversible destruction during preparation is outside this operation. Only the selected design needs physical deployment/preparation. Acquisition and computation happen before the mission unless a separately priced holding mode is supplied.

Because the mean is unbounded as a random loss, an abort cannot simply be charged as an additive total-variation risk. The theorem reports the conditional successful-start mean and the abort probability separately. Alternatively, an uncapped repeat-until-success protocol has a bounded expected preparation overhead, which must be added to the completion mean. A finite hard cap produces an explicit abort outcome.

6.2 Deployment uncertainty

If the actual deployed generator $\widetilde{Q}(c)$ preserves π and its Dirichlet form is between $e^{-\kappa}$ and e^{κ} times that of $Q(c)$ uniformly in c, inverse Loewner order gives

$$e^{-\kappa} M(c) \leq \widetilde{M}(c) \leq e^{\kappa} M(c).$$

If every successful preparation has the above $e^{\pm\beta}$ density envelope, the actual design decision obeys

$$\mathbb{E}_{\nu_{c_T}, \widetilde{Q}(c_T)} \tau_B \leq e^{2(\kappa+\beta)} (1+\varepsilon) \inf_{c \in \mathcal{C}} \mathbb{E}_{\nu_c, \widetilde{Q}(c)} \tau_B.$$

This is a uniform robustness statement, not a finite-observation proof of the multiplier response law. Detector, clock and field errors during acquisition enter C_P's conditional error; actuation calibration enters κ. A directed error breaking detailed balance or changing the invariant law needs a separately proved envelope. It cannot be counted as the same κ by assertion.

7 What this makes possible, and what it cannot make possible

Corollary 4 (No exponential physical speedup from bounded reciprocal gains)**.** *If* $\mathbf{1} \in \mathcal{C}$ *and all feasible coordinates satisfy* $1 \leq c_j \leq C$*, then*

$$\frac{M(\mathbf{1})}{C} \leq M(c) \leq M(\mathbf{1}).$$

Proof. $K(\mathbf{1}) \preceq K(c) \preceq CK(\mathbf{1})$, and inverse order reverses both inequalities in (2). □

An exponentially rare equilibrium target can therefore remain exponentially slow to realize. The capability earned here is global design and experimental decision at a cost independent of that runtime, not free physical realization of the rare product. Exponential gains require exponentially large allowed kinetic gains, a changed invariant, a different initialization, a new transition mechanism, or another resource outside this family. This corollary is a direct consequence of classical form comparison, not a claim of a new universal law of catalysis.

The target family is meaningful beyond a tiny fixture. For N binary variables with bounded-degree interacting Gibbs weights, fixed conditional-surprisal contrast and certified pinned uniqueness, the accepted static construction supplies B as all-terminal disconnection; the present theorem then designs the fastest deliberate stress-test completion within its reciprocal kinetic family. For any singleton target configuration, π_B is deterministic. If all one-site conditional probabilities of its specified bits are at most $q_* < 1$, then $p \le q_*^N$. For sufficiently large N this certifies the rare branch without estimating p at all. Correlated Gibbs interaction and unknown baseline local clocks are allowed under the declared certificates. In both examples $|\Omega| = 2^N$ while the control dimension and oracle reduction are polynomial in local input size and numerical inverse margins.

8 Strongest applicable comparisons and honest scope

Reversible hitting-time variational principles are classical [2]. Convex design of explicit Markov chains, including fastest mixing and effective-resistance objectives, is established [3]; reversible Kemeny optimization has an explicit semidefinite formulation [4]. Inexact Frank–Wolfe and its gap certificate are also established [5]. This work does not claim these ingredients as inventions.

The completed difference is the *succinct unknown-generator interface*: short local experimental updates plus conditional target draws yield a scalar objective oracle with finite confidence, then a global continuum choice, without a matrix with one row per global state, uniform kinetic table recovery, a control grid, or full marked-record certification. An explicit-state SDP has at least exponential input size on the 2^N family; ordinary direct physical hitting-time sampling has exposure proportional to the mean lifetime. These are comparisons of representations under the same supplied local access, not claims that every specialist method is inferior. A method using the same short-return reduction and convex geometry can match this capability.

The accepted whole-CDF machinery already removes inverse-rarity computation in its model. This construction does not claim another exponential rarity speedup over it; it eliminates unnecessary whole-CDF and marked-law outputs and replaces finite-menu enumeration by a global continuous optimization theorem. On independent product models, factorization gives especially strong ordinary integral formulas; the executed diagnostic uses those as an independent reference rather than pretending a general sampler is the best solver for that subclass.

Arbitrary field-dependent clocks invalidate the affine generator family. Finite experiments at finitely many fields cannot certify an unassumed uniform response over a continuum. Maximizing the convex objective for catastrophe prevention is a different optimization problem. Adaptive feedback policies, all-working initial states, irreversible setup, unknown hidden mechanisms, nonreversible driving, and changing invariant laws are not supplied by this result.

Proposition 5 (Why the response premise cannot be replaced by finite queries)**.** *Consider a one-bit model at fields $u \in [0,1]$ with fully known logit u and rates $r(u)\sigma(u)$ from zero and $r(u)[1-\sigma(u)]$ from one. Let r range over smooth functions in $[1, C]$, with no supplied modulus of continuity or derivative bound. Even an oracle that returns the exact $r(u)$ at each queried field cannot, with a uniformly bounded number of queries, guarantee a global constant-factor approximation to minimum mean completion time over this class when the desired factor is smaller than $C\sigma(0)/\sigma(1)$. This is a finite-query statement, not a lower bound for an unpriced infinitely resolving continuous sweep.*

Proof. Place m disjoint open intervals in $[0,1]$. In instance j, set $r_j(u) = 1 + (C-1)b_j(u)$ where b_j is a smooth bump supported in interval j and equal to one at its center. Until an algorithm queries inside the selected interval, its transcript is identical to that for $r = 1$. For a uniform random j, a randomized algorithm making at most q queries and outputting one field intersects the selected interval with probability at most $(q+1)/m$: follow its transcript on the baseline, then count the intervals containing one of its queries or its output. If it misses, its completion rate is at most $\sigma(1)$ whereas the optimum is at least $C\sigma(0)$. Hence its mean-time approximation factor is at least $C\sigma(0)/\sigma(1)$. Letting m grow defeats any uniform finite query bound. Faithful finite experiments are no stronger than this exact response oracle. □

The known affine multiplier law in (1) is therefore a substantive structural intervention resource. A finite calibration with a proved response modulus can instead be charged at that modulus's scale; the proposition does not forbid such a construction.

9 Execution record

The accompanying code checks the objective and derivative identities on small independent chains by two representations (explicit killed linear algebra and a product integral), and performs continuum allocation on a growing independent family using the strongest available factorized integral as an audit oracle. It records numerical optimization gaps and target rarity. These are deterministic mathematical diagnostics under known synthetic parameters. They do not execute the general correlated static service, the conservative finite experimental schedule, a physical kinetic multiplier, or a real rare completion. Details and any failed checks are retained in the execution record. Numerical quadrature errors are separately reported; those floating diagnostics are not a substitute for the finite rational theorem.

N	Target mass p	Relative numerical FW gap	Mean after design
8	$8.78 \cdot 10^{-3}$	$3.92 \cdot 10^{-5}$	18.02
16	$1.25 \cdot 10^{-4}$	$1.19 \cdot 10^{-5}$	516.76
32	$7.77 \cdot 10^{-9}$	$1.99 \cdot 10^{-5}$	$3.99 \cdot 10^{6}$
64	$1.22 \cdot 10^{-16}$	$1.63 \cdot 10^{-5}$	$1.21 \cdot 10^{14}$
128	$5.33 \cdot 10^{-32}$	$4.35 \cdot 10^{-6}$	$1.36 \cdot 10^{29}$

The state space at $N = 128$ has 2^{128} states. The improvement over equal feasible allocation ranges only from 1.096 to 1.233 across these examples, as expected under bounded kinetic gains. An ordinary constrained optimizer on the same product integral agrees within the numerical gap; this is not a separation from that strong subclass solver. Six small explicit killed-matrix checks agree with the product formula and its derivative homogeneity identity. A separate interacting three-bit 90-decimal-digit check has $p = 3.13 \cdot 10^{-16}$ and $k = 1242$; $f = 4.022433879193464\ldots$ and $1/z = 4.022433879193855\ldots$, an error $3.90 \cdot 10^{-13}$ below the declared allowance 0.005.

The sufficient general experimental schedule is *not* the fast product calculation. For the $N = 128$ instance, $\varepsilon = 0.1$ and $\delta = 0.05$, the elementary theorem schedule gives approximately $4.12 \cdot 10^{8}$ scalar calls at $\nu = 7.46 \cdot 10^{-15}$ and $10^{61.65}$ potential operating updates before the per-update physical factory bill. It is evaluated, not executed. The absence of an inverse global-rarity term is therefore a theoretical scaling result, not an earned useful experimental regime. Neither these constants nor this schedule are lower bounds. A practical implementation would need a different gradient/oracle realization or stronger instance structure; that improvement is not claimed here.

References

[1] Preserved Astra acquired first-event package, `acquired_first_event_research_package`, including `next5_decision/experimental_operating.tex`, `experimental_static.tex`, and the preceding finite CDF proof. Every file is retained unchanged; hashes are in the companion record.

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